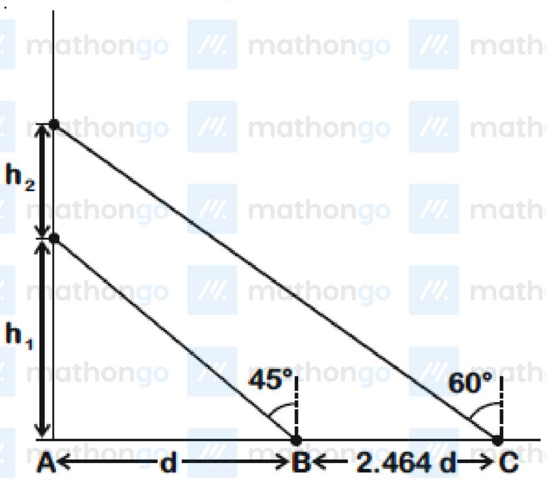
Q1 JEE Main 2020 - 4 September (Morning)

Starting from the origin at time $t=0$, with initial velocity $5\hat{\mathrm{j}}\,\mathrm{ms}^{-1}$, a particle moves in the $\mathrm{x}-\mathrm{y}$ plane with a constant acceleration of $(10\hat{\mathrm{i}}+4\hat{\mathrm{j}})\mathrm{ms}^{-2}$. At time t, its coordinates are (20 $\mathrm{m}, \mathrm{y}_0\mathrm{m})$. The values of t and y_0 are, respectively

(A) $4\mathrm{s}$ and $52\mathrm{m}$
(B) $2\mathrm{s}$ and $24\mathrm{m}$
(C) $5\mathrm{s}$ and $25\mathrm{m}$
(D) $2\mathrm{s}$ and $18\mathrm{m}$

Q2 JEE Main 2020 - 5 September (Morning)

A balloon is moving up in air vertically above a point A on the ground. When it is at a height h_1 a girl standing at a distance d (point B) from A (see figure) sees it at an angle 45° with respect to the vertical. When the balloon climbs up a further height h_2, it is seen at an angle 60° with respect to the vertical if the girl moves further by a distance $2.464\mathrm{d}$ (point C). Then the height h_2 is (given $\tan 30^{\circ}=0.5774)$



(A) d
(B) $0.732\mathrm{d}$
(C) $1.464\mathrm{d}$
(D) $0.464\mathrm{d}$

Q3 JEE Main 2020 - 6 September (Morning)

A clock has a continuously moving second's hand of $0.1\mathrm{m}$ length. The average acceleration of the tip of the hand (in units of ms^{-2}) is of the order of
(A) 10^{-1}
(B) 10^{-2}
(C) 10^{-4}
(D) 10^{-3}

Q4 JEE Main 2020 - 6 September (Evening)

When a car is at rest, its driver sees rain drops falling on it vertically. When driving the car with speed v, he sees that rain drops are coming at an angle 60° from the horizontal. On further increasing the speed of the car to $(1+\beta)v$, this angle changes to 45°. The value of β is close to

(A) 0.37
(B) 0.41
(C) 0.73
(D) 0.5

Q5 JEE Main 2020 - 8 January (Evening)

A particle moves such that its position vector $\vec{r}(t)=cos\omega t\ \hat{i}+\sin\ \omega t\ \hat{j}$ where ω is a constant and t is time. Then which of the following statements is true for the velocity $\vec{v}(t)$ and acceleration $\vec{a}(t)$ of the particle :

(A) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is towards origin
(B) $\vec{v}$ and $\vec{a}$ are perpendicular to $\vec{r}$
(C) $\vec{v}$ is parallel to $\vec{r}$ and $\vec{a}$ parallel to $\vec{r}$.
(D) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is away from origin.

Q6 JEE Main 2020 - 9 January (Evening)

A particle starts from the origin at $t=0$ with an initial velocity of $3.0\ \hat{i}\ m/s$ and moves in the x-y plane with a constant acceleration $\left(6.0\hat{i}+4.0\hat{j}\right)\ m/s^2$. The x-coordinate of the particle at the instant when its y-coordinate is $32\ m$ is D meters. The value of D is :
(A) 50

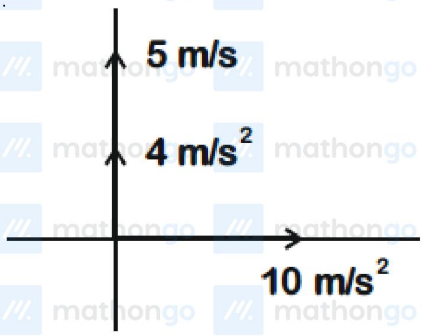
(B) 40
(C) 32
(D) 60

Answer Key

Q1 (D) **Q2** (A) **Q3** (D) **Q4** (C)
Q5 (A) **Q6** (D)

Q1

$Att = t, (20m, y_0)$



$$\therefore t = \sqrt{\frac{20 \times 2}{10}}$$
$$= 2\mathrm{s}$$
$$\therefore \quad y_0 = 5 \times 2 + \frac{1}{2} \times 4 \times 2^2$$
$$= 18\mathrm{m}$$

Q2

$\frac{h_1}{d} = \tan 45^\circ$

$\frac{\mathrm{h_1 + h_2}}{\mathrm{d + 2.464d}} = \tan 30^\circ$

$\Rightarrow \mathrm{h_2 = d}$

Q3

$a = \omega^2 \times \ell$

$= \left(\frac{2\pi}{\mathrm{T}}\right)^2 \times \ell$

$= \left(\frac{2\pi}{60}\right)^2 \times 0.1$

$= 1.1 \times 10^{-3} \mathrm{m/s^2}$

Q4

Given

$\frac{v^1}{v} = \tan 60 \ldots \text{(i)} \longrightarrow$ 

$\frac{v^1}{v(1+\beta)} = \tan 45^\circ \quad \ldots \text{(ii)} \quad \frac{}{v(1+\beta)}$ 

From (i) and (ii)

$\beta = 0.73$

Q5

$$\vec{r} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$$
$$\vec{v} = \frac{d\vec{r}}{dt} = \omega\left(-\sin \omega t \hat{i} + \cos \omega t \hat{j}\right)$$
$$\vec{a} = \frac{d\vec{v}}{dt} = -\omega^2\left(\cos \omega \widehat{t i} + \sin \omega t \hat{j}\right)$$
$$\vec{a} = -\omega^2 \vec{r}$$
$$\therefore \quad \vec{a} \text{ is antiparallel to } \vec{r}$$
$$\vec{v} \cdot \vec{r} = \omega\left(-\sin \omega t \cos \omega t + \cos \omega t \sin \omega t\right) = 0$$
$$\text{so } \vec{v} \perp \vec{r}$$

Q6

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$32 = 0 + \frac{1}{2} \times 4t^2 \quad \Rightarrow \quad t = 4 \sec$$

$$S_x = u_x t + \frac{1}{2} a_x t^2$$

$$= 3 \times 4 + \frac{1}{2} \times 6 \times 16$$

$$= 60\ m.$$